

Artavazd Maranjyan

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Current Position

2026–present **Postdoctoral Researcher, working with Volkan Cevher and Martin Jaggi, EPFL, Lausanne, Switzerland**

Education

2023–2025 **Ph.D. in Computer Science, KAUST, Thuwal, Saudi Arabia; Advisor:** Peter Richtárik

📄 **Thesis:** [First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data](#)

2021–2023 **M.Sc. in Applied Statistics and Data Science, Yerevan State University, Yerevan, Armenia**

📄 **Thesis:** [On local training methods](#); **Co-supervisors:** Peter Richtárik, Mher Safaryan

2017–2021 **B.Sc. in Informatics and Applied Mathematics, Yerevan State University, Yerevan, Armenia**

📄 **Thesis:** [On the Convergence of Series in Classical Systems](#); **Supervisor:** Martin Grigoryan

Academic Experience

03/2025 **Research Visit to Yi-Shuai Niu, BIMSA, Beijing, China**

- Gave talks at three universities (PKU, BUAA, BIMSA)
- Collaborated with Professor Yi-Shuai Niu on a project on Server-Assisted Federated Learning

04/23–08/23 **Researcher in the Group of Martin Grigoryan, Yerevan State University, Yerevan, Armenia**

- Studied the existence and properties of universal functions with respect to the Vilenkin and Haar systems across various functional spaces

06/22–01/23 **Internship in the Group of Peter Richtárik, KAUST, Thuwal, Saudi Arabia**

- Worked on the "GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity" paper

01/22–08/23 **Machine Learning Researcher, YerevaNN, Yerevan, Armenia**

- Worked on the intersection of Federated Learning and Optimization

Industry Experience

07/21–06/22 **Co-Founder, OnePick, Yerevan, Armenia**

- Built an MVP platform to help content creators and marketers generate or select photos that best matched their text descriptions
- Participated in the [InVent 2.0 \(FAST\)](#) venture building program, advancing through team formation, idea generation, product development, and pitching
- Secured initial funding and continued development beyond the program

07/21–09/21 **Backend Developer, EXALT Technologies Ltd, Yerevan, Armenia**

- Worked for Nutanix.

06/21–07/21 **Machine Learning Research Engineer, FAST, Yerevan, Armenia**

- Worked on Fraud detection
- Made data-driven forecasts using machine learning algorithms and statistical models

09/19–01/21 **Software Engineer in Test, Picsart, Yerevan, Armenia**

- Worked with automation team to design and develop automated solutions across several mobile/web applications
- Worked directly with software developers, test engineers, product owners, business analysts to find and resolve issues
- Worked closely with DevOps to suggest improvements in processes and in Jenkins Continuous Integration cycle

Awards

06/2026 **EPFL AI Center and Swiss AI Initiative Postdoctoral Fellowships, EPFL**

Competitive fellowship supporting independent postdoctoral research, with additional travel funding

- 05/2025 **CEMSE Dean's List Award, KAUST**
Awarded for excellent academic and research performance (\$2,500 prize)
- 09/2023 **Dean's Award, KAUST**
Awarded to a few top students accepted to KAUST (\$6,000 annually for 3 years)
- 05/2021 **Outstanding Final Project Award, Yerevan State University**
Recognized for the Bachelor's thesis (awarded to 6 students among 250+ students)

Publications

15. **LOSCAR-SGD: Local SGD with Communication-Computation Overlap and Delay-Corrected Sparse Model Averaging**
Yassine Maziane, Ammar Mahran, [Artavazd Maranjyan](#), Peter Richtárik
arXiv:2605.20866, 2026
14. **Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method**
Abdurakhmon Sadiev, [Artavazd Maranjyan](#), Ivan Ilin, Peter Richtárik
arXiv:2605.18174, 2026
13. **Rescaled Asynchronous SGD: Optimal Distributed Optimization under Data and System Heterogeneity**
Ammar Mahran, [Artavazd Maranjyan](#), Peter Richtárik
arXiv:2605.13434, 2026
12. **Rennala MVR: Improved Time Complexity for Parallel Stochastic Optimization via Momentum-Based Variance Reduction**
Zhirayr Tovmasyan, [Artavazd Maranjyan](#), Peter Richtárik
arXiv:2605.08871, 2026
11. **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity**
[Artavazd Maranjyan](#), Peter Richtárik
ICLR 2026: The Fourteenth International Conference on Learning Representations
10. **BiCoLoR: Communication-Efficient Optimization with Bidirectional Compression and Local Training**
Laurent Condat, [Artavazd Maranjyan](#), Peter Richtárik
arXiv:2601.12400, 2026
9. **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning**
[Artavazd Maranjyan](#), El Mehdi Saad, Peter Richtárik, Francesco Orabona
ICML 2025: Forty-Second International Conference on Machine Learning
8. **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity**
[Artavazd Maranjyan](#), Alexander Tyurin, Peter Richtárik
ICML 2025: Forty-Second International Conference on Machine Learning
7. **MindFlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times**
[Artavazd Maranjyan](#), Omar Shaikh Omar, Peter Richtárik
UAI 2025: The 41st Conference on Uncertainty in Artificial Intelligence
OPT 2024: Optimization for Machine Learning (NeurIPS workshop)
Oral presentation (top 5% of 107 submissions)
6. **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity**
[Artavazd Maranjyan](#), Mher Safaryan, Peter Richtárik
TMLR 2025: Transactions on Machine Learning Research
5. **LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression**
Laurent Condat, [Artavazd Maranjyan](#), Peter Richtárik
ICLR 2025: The Thirteenth International Conference on Learning Representations
Spotlight presentation (top 5.1% of the submitted papers)

4. **Differentially Private Random Block Coordinate Descent**
Artavazd Maranjyan, Abdurakhmon Sadiev, Peter Richtárik
OPT 2024: Optimization for Machine Learning (NeurIPS workshop)
3. **Menshov-type theorem for divergence sets of sequences of localized operators**
Martin Grigoryan, Anna Kamont, Artavazd Maranjyan
Journal of Contemporary Mathematical Analysis, vol. 58, no. 2, pp. 81-92, 2023
2. **On the divergence of Fourier series in the general Haar system**
Martin Grigoryan, Artavazd Maranjyan
Armenian Journal of Mathematics, vol. 13, pp. 1-10, 2021
1. **On the unconditional convergence of Faber-Schauder series in L^1**
Tigran Grigoryan, Artavazd Maranjyan
Proceedings of the YSU A: Physical and Mathematical Sciences, vol. 55, no. 1 (254), pp. 12-19, 2021

Reviewing

- 2024–present TMLR (Transactions on Machine Learning Research)
- 2026 NeurIPS (Conference on Neural Information Processing Systems)
- 2026 ICML (International Conference on Machine Learning)
- 2025 ICLR (International Conference on Learning Representations)
- 2024 SIMODS (SIAM Journal on Mathematics of Data Science)
- 2024 JMLR (Journal of Machine Learning Research)

Talks and Poster Presentations

- 26/04/2026 **Protocol Learning: Decentralized Collaborative Learning at Scale (ICLR 2026 Workshop)**, Lagune Barra Hotel, Rio de Janeiro, Brazil
Delivered a talk titled **Controlling Delay in Asynchronous SGD: Optimality for Any Data Regime** [\[video\]](#) [\[slides\]](#)
- 23/04/2026 **(ICLR 2026) The Fourteenth International Conference on Learning Representations**, Riocentro Convention and Event Center, Rio de Janeiro, Brazil
Presented a poster on **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity** [\[poster\]](#)
- 09/02/2026 **KAUST Rising Stars in AI Symposium 2026**, KAUST, Thuwal, Saudi Arabia
Presented posters on
 - **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity** [\[poster\]](#)
 - **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[poster\]](#)
 - **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning** [\[poster\]](#)
- 13/11/2025 **AMCS/STAT Graduate Seminar**, KAUST, Thuwal, Saudi Arabia
Delivered a talk titled **First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data** [\[slides\]](#)
- 04/11/2025 **The KAUST 2025 Workshop on Statistics**, KAUST, Thuwal, Saudi Arabia
Presented posters on
 - **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[poster\]](#)
 - **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning** [\[poster\]](#)
- 28/10/2025 **Mathematics and Applications Colloquium**, KAUST, Thuwal, Saudi Arabia
Delivered a talk on **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity** [\[slides\]](#)
- 29/07/2025 **35º Colóquio Brasileiro de Matemática**, IMPA, Rio de Janeiro, Brazil
Delivered a talk and a poster on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [\[slides\]](#) [\[poster\]](#)
- 23/07/2025 **41st Conference on Uncertainty in Artificial Intelligence**, Rio Othon Palace, Rio de Janeiro, Brazil
Presented a poster on **MindPlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times** [\[poster\]](#)

- 16/07/2025 **Forty-Second International Conference on Machine Learning**, Vancouver Convention Center, Canada
Presented posters on
- Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [poster]
 - ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning [poster]
- 18/05/2025 **Stochastic Numerics and Statistical Learning**, KAUST, Thuwal, Saudi Arabia
Presented a poster on ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning [poster]
- 26/04/2025 **The Thirteenth International Conference on Learning Representations**, Singapore EXPO, Singapore
Presented a poster on LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression (Spotlight presentation (top 5.1% of the submitted papers))
- 16/04/2025 **FLOW Talk #126**, Federated Learning One World Seminar (FLOW), Online
Delivered a talk on Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [video] [slides]
- 08/04/2025 **KAUST Rising Stars in AI Symposium 2025**, KAUST, Thuwal, Saudi Arabia
Presented a poster on ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning [poster]
- 28/03/2025 **Machine Learning Reading Group**, YSU Krisp-AI Lab, Yerevan, Armenia
Delivered a talk on Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [slides]
- 26/03/2025 **Flower AI Summit 2025**, King's House, London, England
Delivered a talk on Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [slides]
- 17/03/2025 **Academic Report of the School of Mathematical Sciences**, Beihang University (BUAA), Beijing, China
Invited by Jiaxin Xie to give a talk on Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [slides]
- 13/03/2025 **Optimization Seminar**, BIMSA, Beijing, China
Invited by Yi-Shuai Niu to give a talk on Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [slides]
- 12/03/2025 **Seminar**, Peking University, Beijing, China
Invited by Kun Yuan to give a talk on Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [slides]
- 27/02/2025 **AMCS/STAT Graduate Seminar**, KAUST, Thuwal, Saudi Arabia
Delivered a talk on Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity [slides]
- 15/12/2024 **Workshop on Optimization for Machine Learning (NeurIPS 2024)**, Vancouver Convention Center, Vancouver, Canada
Presented
- MindFlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times (Oral presentation, top 5% of 107 submissions) [video]
 - Differentially Private Random Block Coordinate Descent
 - LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression
- 21/11/2024 **MLR Weekly Seminar**, Machine Learning Research at Apple, Online
Invited by Samy Bengio to give a talk on MindFlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times [slides]
- 18/10/2024 **International Conference on Algebra, Logic, and their Applications**, Yerevan State University, Online
Delivered a talk on MindFlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times
- 10/10/2024 **CEMSE E-Poster Competition**, KAUST, Thuwal, Saudi Arabia
Awarded 3rd place for presenting a poster on GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity
- 06/07/2024 **Analysis, PDEs and Applications**, Yerevan State University, Yerevan, Armenia
Delivered a talk on MindFlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times [abstract]

- 27/05/2024 **Stochastic Numerics and Statistical Learning**, KAUST, Thuwal, Saudi Arabia
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [poster]
- 05/05/2024 **CS 331: Stochastic Gradient Descent Methods**, KAUST, Thuwal, Saudi Arabia
Delivered a guest lecture on **MindFlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times**
- 13/03/2024 **The Machine Learning Summer School in Okinawa 2024**, OIST, Okinawa, Japan
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [poster]
- 21/02/2024 **KAUST Rising Stars in AI Symposium 2024**, KAUST, Thuwal, Saudi Arabia
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [poster]
- 16/11/2023 **Group Seminar**, KAUST, Thuwal, Saudi Arabia
Delivered a talk on **Differentially Private Coordinate Descent for Composite Empirical Risk Minimization**
- 06/10/2023 **Algorithms & Computationally Intensive Inference seminars**, University of Warwick, Coventry, England
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [slides]
- 05/07/2023 **Mathematics in Armenia: Advances and Perspectives**, Yerevan State University, Yerevan, Armenia
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [abstract]
- 10/03/2023 **Machine Learning Reading Group**, Yerevan State University, Yerevan, Armenia
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [video (Yerevan, Armenia)]
- 07/12/2022 **FLOW Talk #88**, Federated Learning One World Seminar (FLOW), Online
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [video]
- 10/04/2022 **Machine Learning Reading Group**, Yerevan State University, Yerevan, Armenia
Delivered a talk on **ProxSkip: Yes! Local Gradient Steps Provably Lead to Communication Acceleration! Finally!**